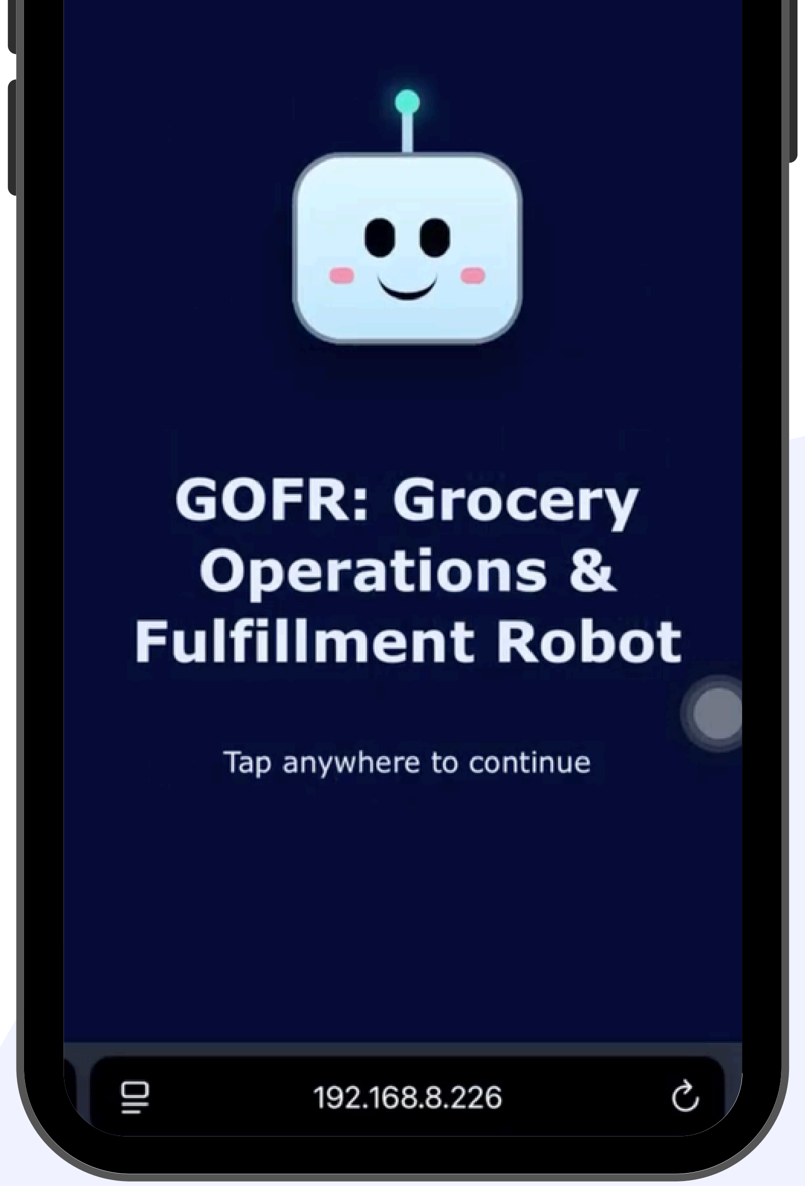
SICK $10K Challenge 2025-26

GOFR

GROCERY OPERATIONS & FULFILLMENT ROBOT



BOSTON UNIVERSITY

Boston University College of Engineering

Revolutionizing Grocery Store Automation

MEET OUR TEAM

Professor Thomas Little
Electrical & Computer Engineering

ADVISOR

Xingjian Jiang

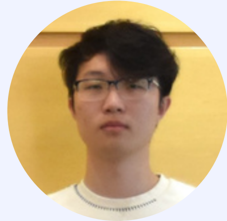
Electrical & Computer Engineering

01

Pree Simphliphan

Electrical & Computer Engineering

02



Feng Tai

Electrical & Computer Engineering

03

Darren Figo Sajino

Mechanical Engineering

04

Bach Thien Nguyen

Mechanical Engineering

05

Bernie Xu

Mechanical Engineering

06

GOFR - Grocery Operations & Fulfillment Robot

PROBLEM STATEMENT

The grocery retail industry faces critical operational challenges that impact efficiency, profitability, and customer satisfaction. These growing pain points demand innovative solutions:

- **Severe labor shortages**
- **Rising operational and staffing costs**
- **Inefficient manual restocking processes**
- **Growing customer demand for online ordering**

To address these challenges, GOFR provides an autonomous robotic solution that streamlines grocery operations, reduces labor dependency, and enables seamless online order fulfillment.

PROPOSED SOLUTION

Autonomous Robot Solution

Deploy GOFR robots for automated restocking and order fulfillment in grocery stores in a single robot, equipped with the SICK picoscan150 LiDAR, offering real-time mapping, autonomous navigation with integration with robotics arm and camera for product detection while providing safety features

Store Manager App Integration

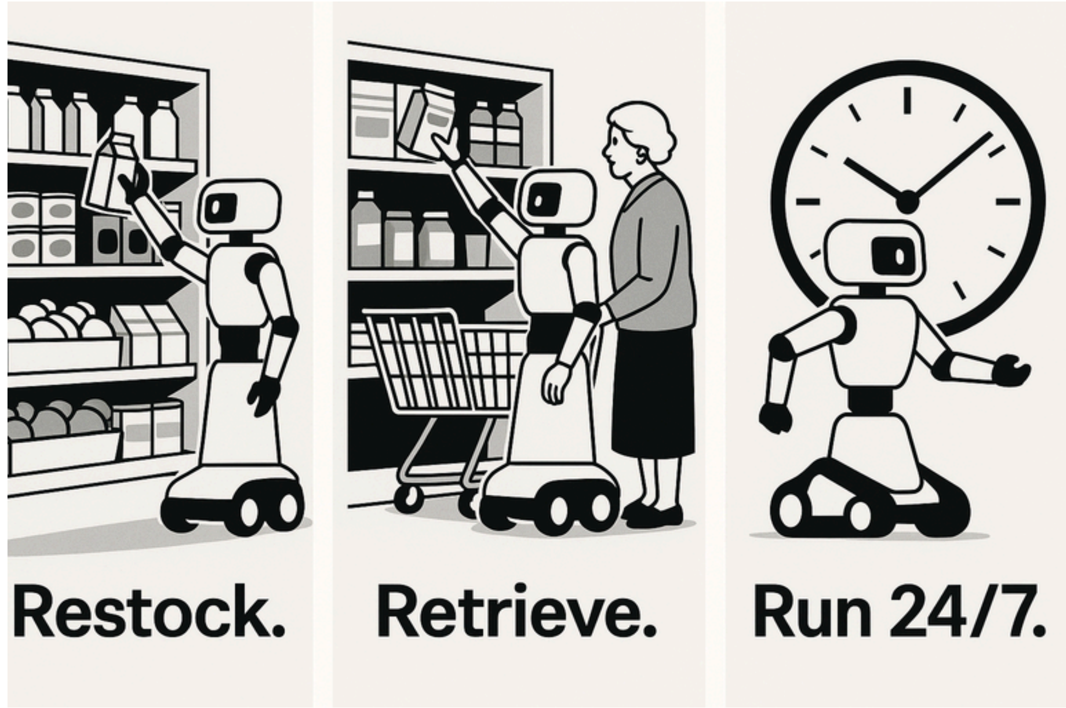
Enable store managers to monitor inventory, AI analyze data, store map reassignment, assign tasks, robot status check, and remotely control robot operations.

Customer App

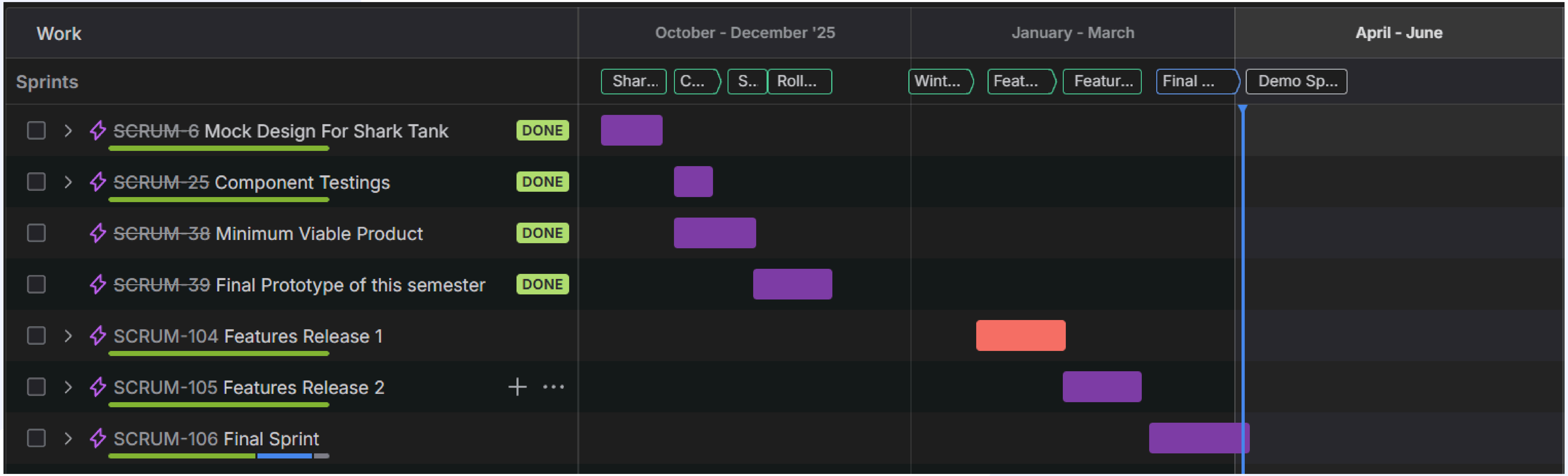
Allow customers to place online orders fulfilled by GOFR robots

24/7 Operations

Improve efficiency with 24/7 autonomous grocery operations while cooperating with workers during daytime



PROJECT PLAN



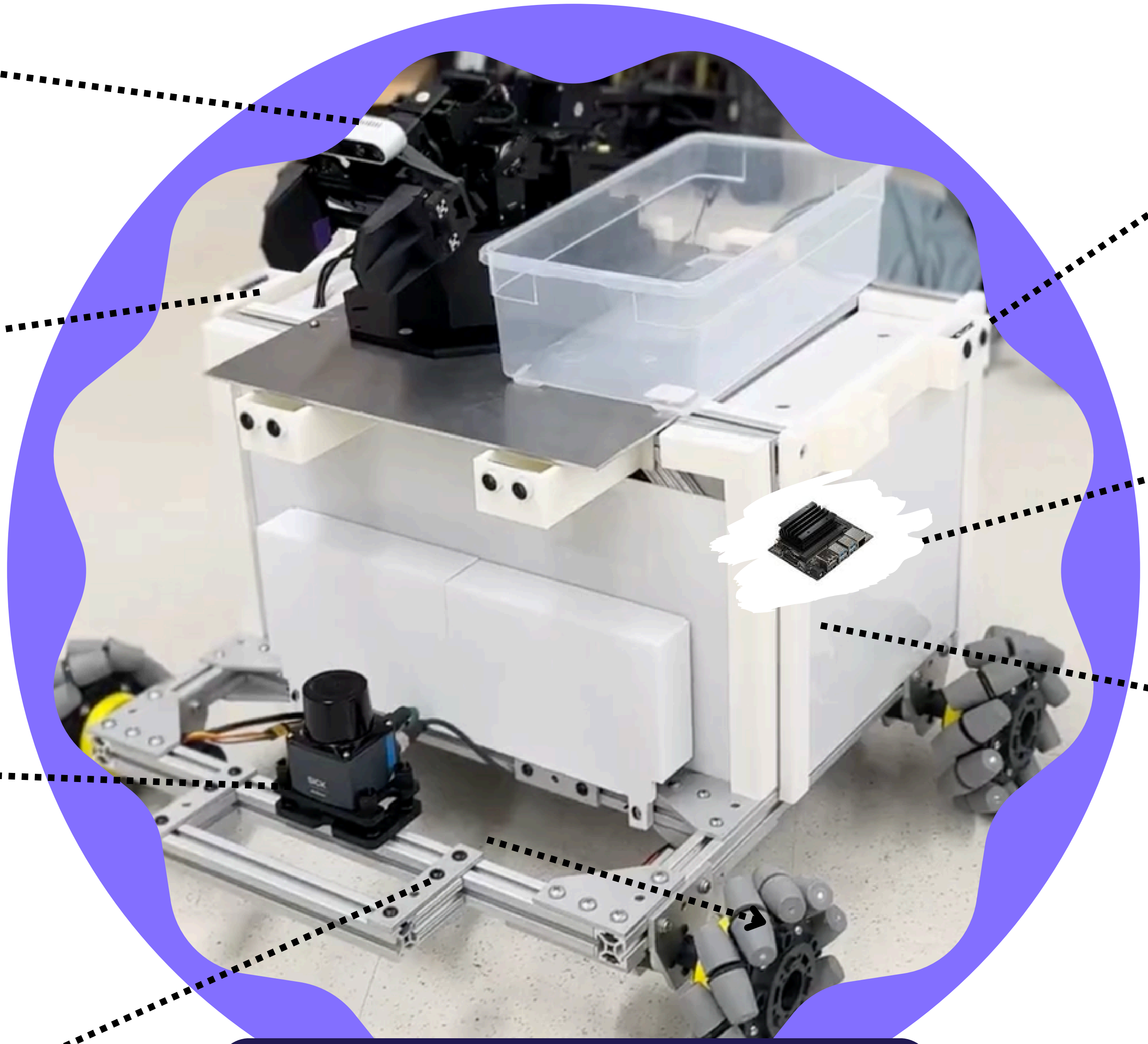
ROBOT PROTOTYPE

IntelRealsense Camera for product recognition

ViperX 300S Robotic Arm for product pickup

SICK picoscan150 LiDAR for autonomous mapping and navigation

80/20 aluminum extrusion for strong structure



More Documentation Detail, visit our github repo:

https://github.com/preespp/EC463_Team_21_Grocery_Robot

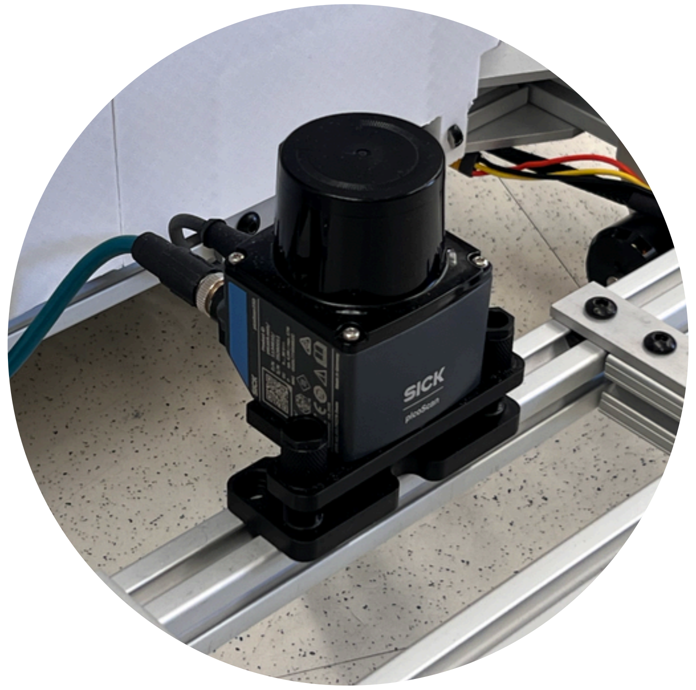
Ultrasonic Sensors + Remote-Estop for 2^{nd} + 3^{rd} safety layer in addition to software stop

NVIDIA Jetson Nano for high computational execution

Acrylic Plates + PETG 3D-printed Casings for lightweight, non-toxic, and protection

Mecanum Wheels for smooth omnidirectional movement

CORE TECHNOLOGY



Tech Stack

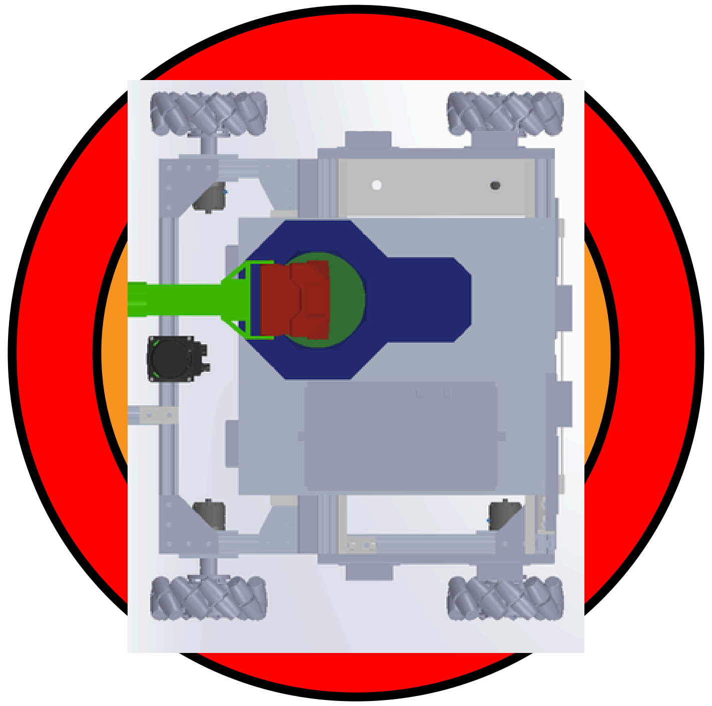
N A V 2

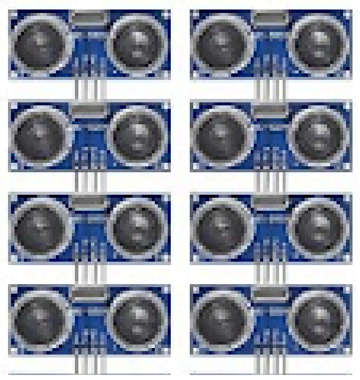
cartographer

Picoscan150 LiDAR Navigation System

360° environmental scanning for precise mapping and real-time positioning in store aisles. We overlay a semantic map that tags location slots, allowing the robot to convert high-level goals like "go to milk" into precise navigation and service poses.

01





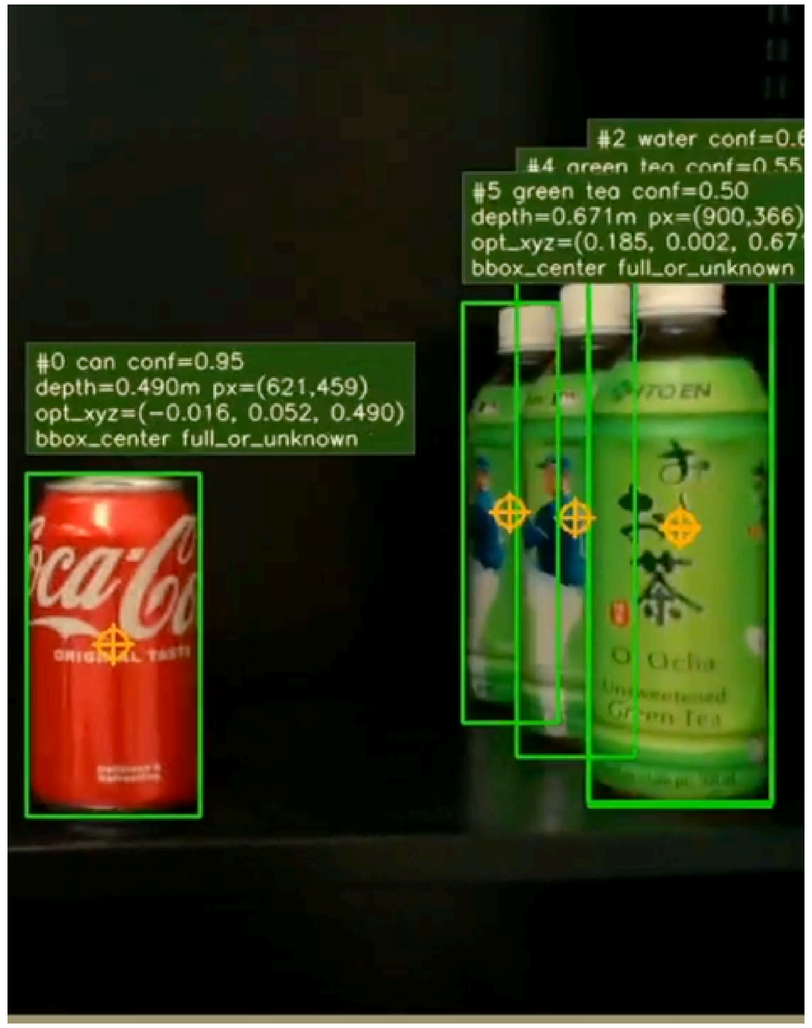
Safety features

We have 3 safety features.

1. Remote Emergency Stop (Relay)
2. Software Logic Avoidance
3. Ultrasonic Sensor 360° Collision Avoidance

02

CORE TECHNOLOGY



Camera + Computer Vision

Our pipeline provides semantic perception: it identifies products, estimates their positions relative to the robot, and supports picking, restocking, and target verification. Computer vision tells the robot what object it is looking at and where it is on the shelf.

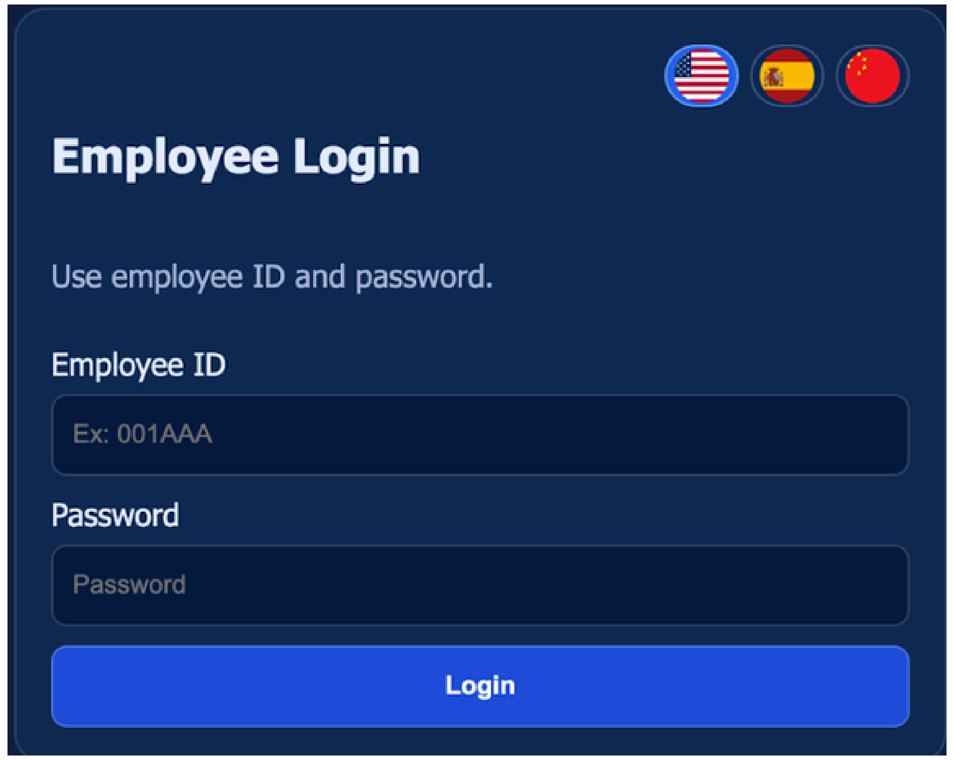
03

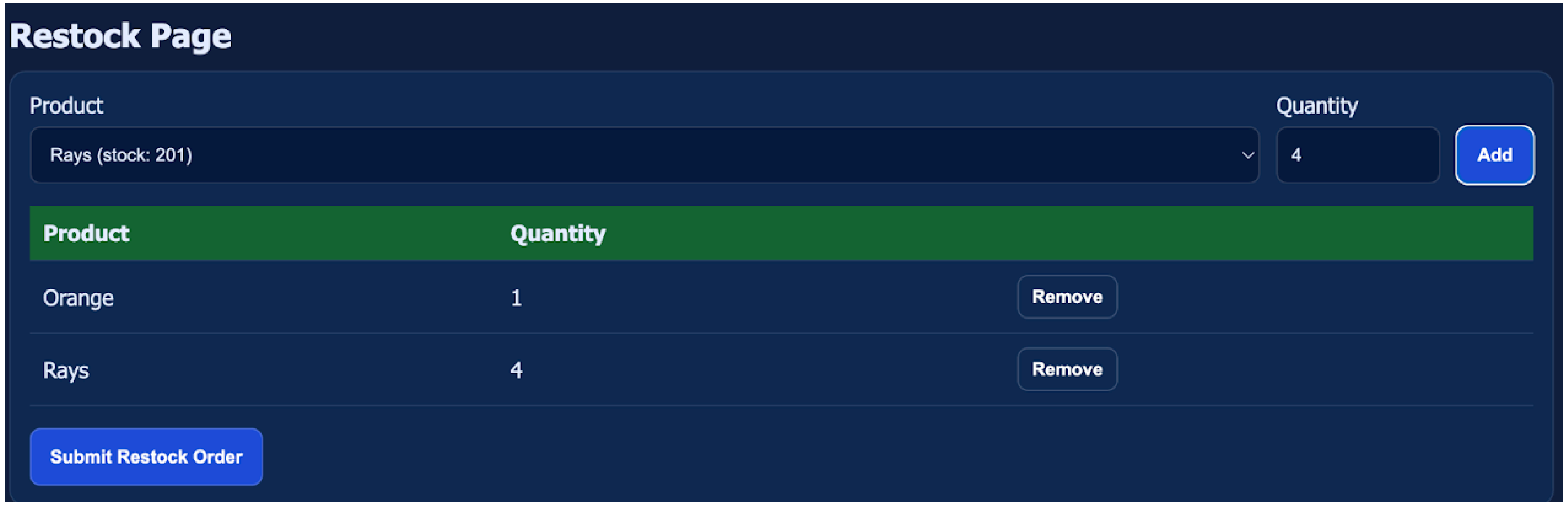
ViperX 300S Robotic Arm Mechanism

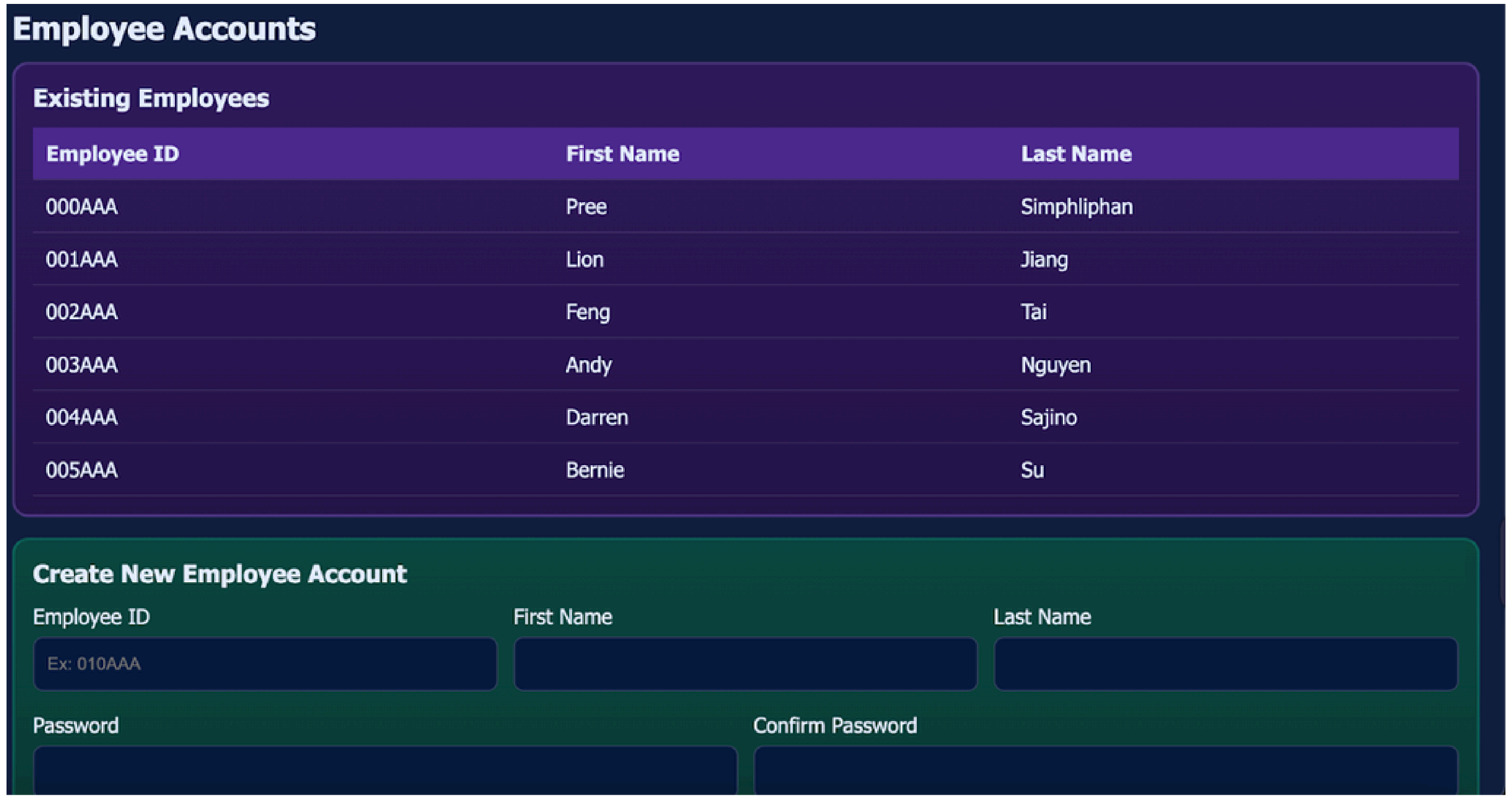
Articulated gripper system for picking, placing, and restocking products of various sizes.

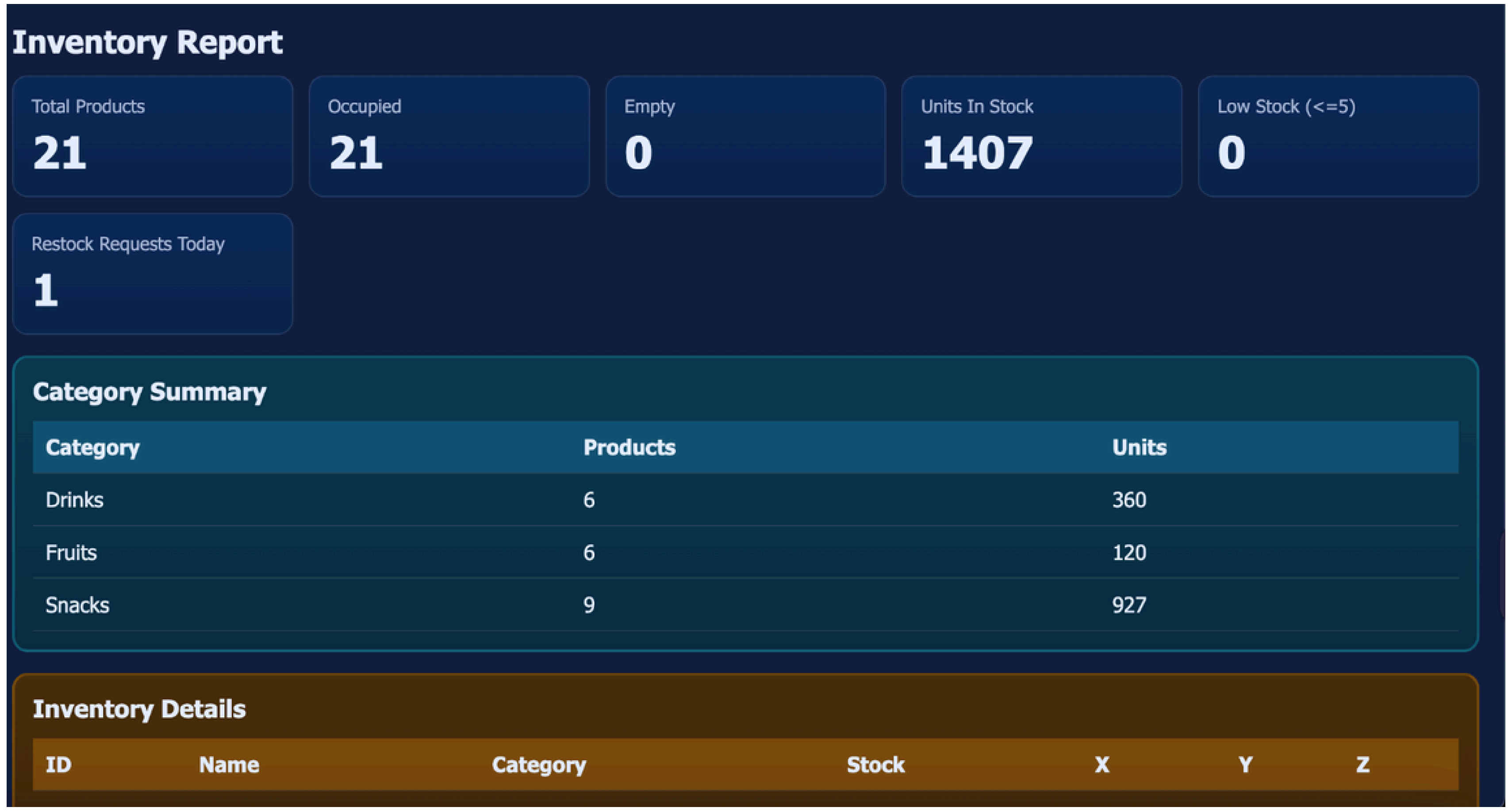
04

05 STORE MANAGER APP

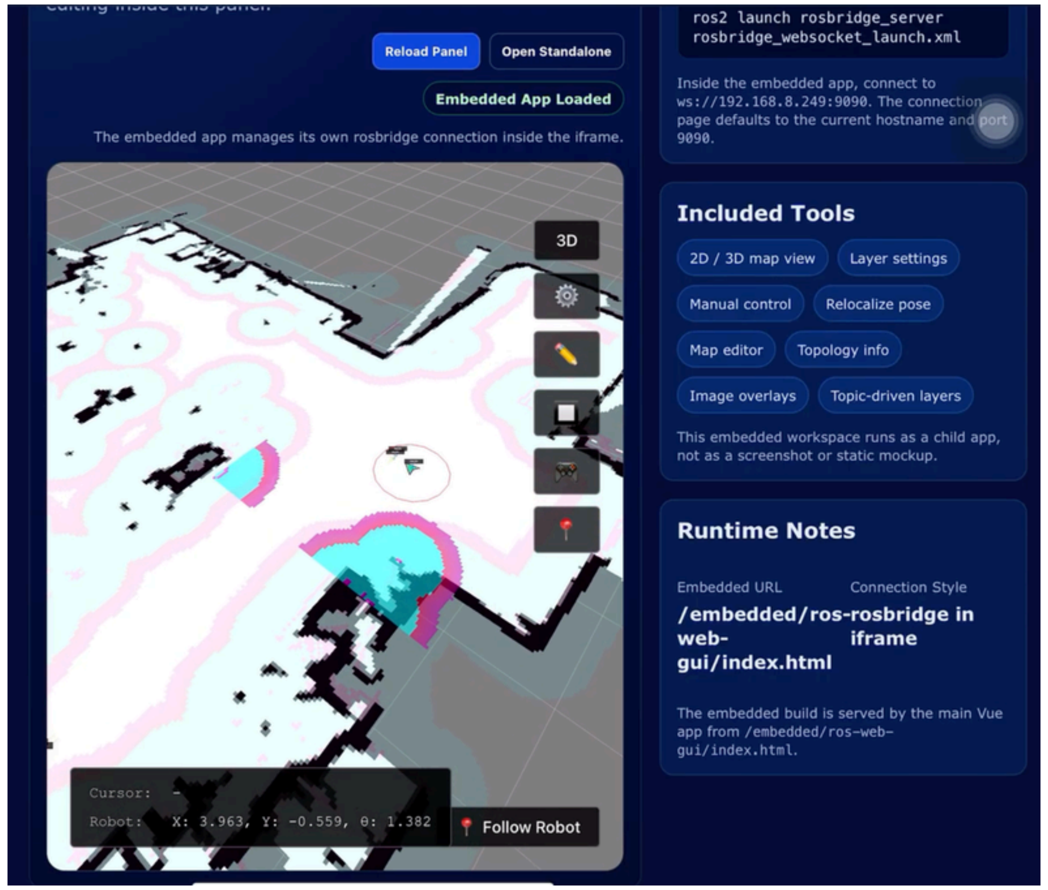






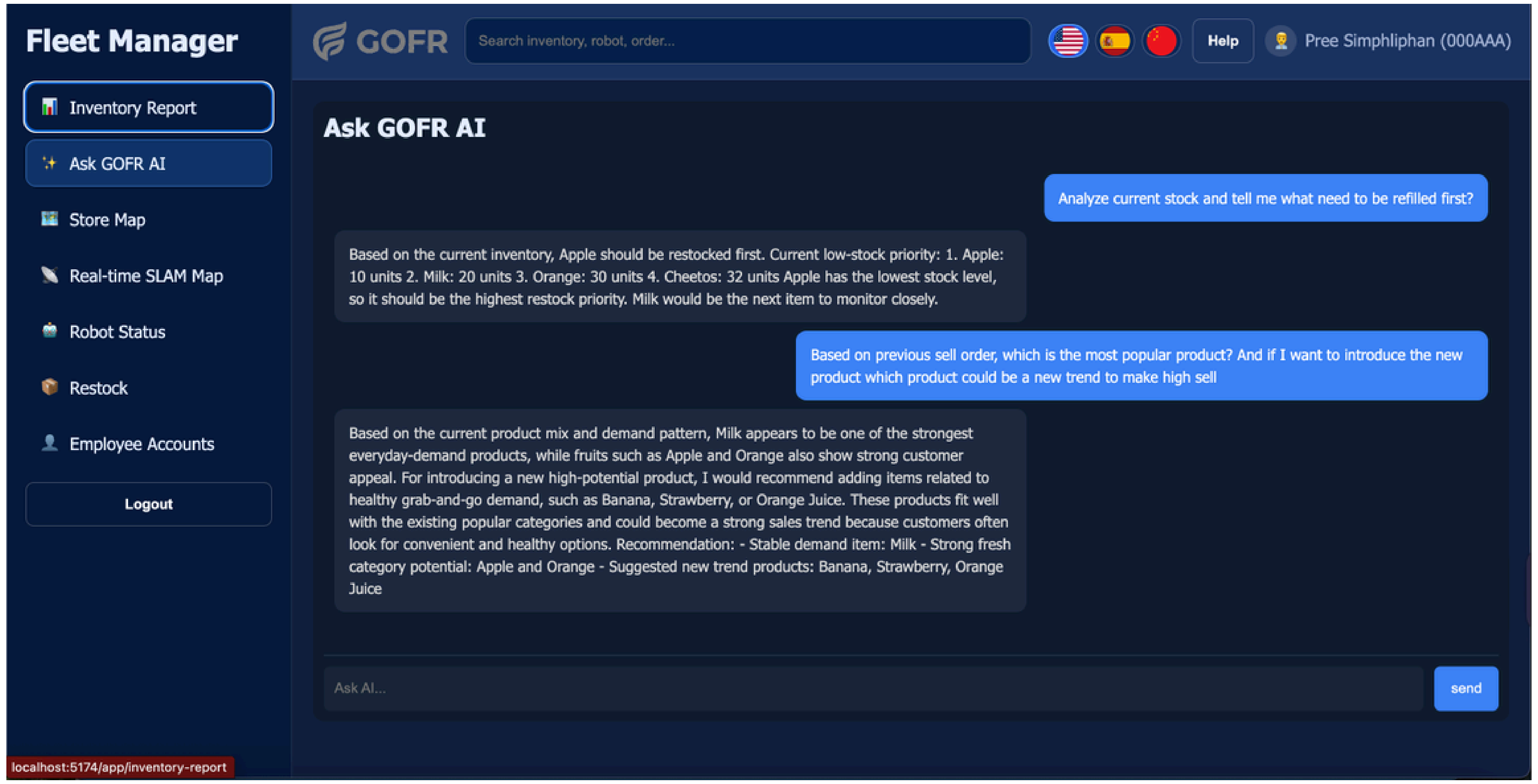


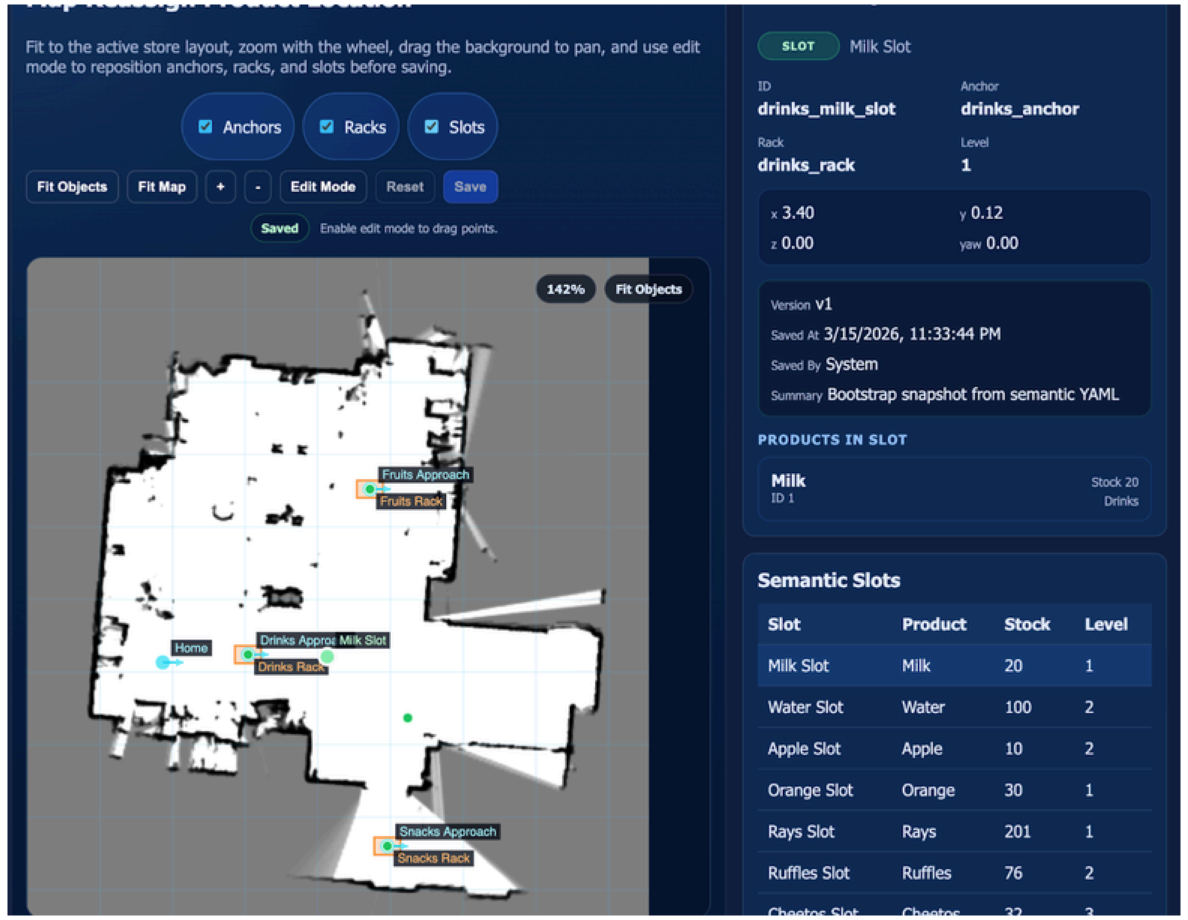
05 STORE MANAGER APP



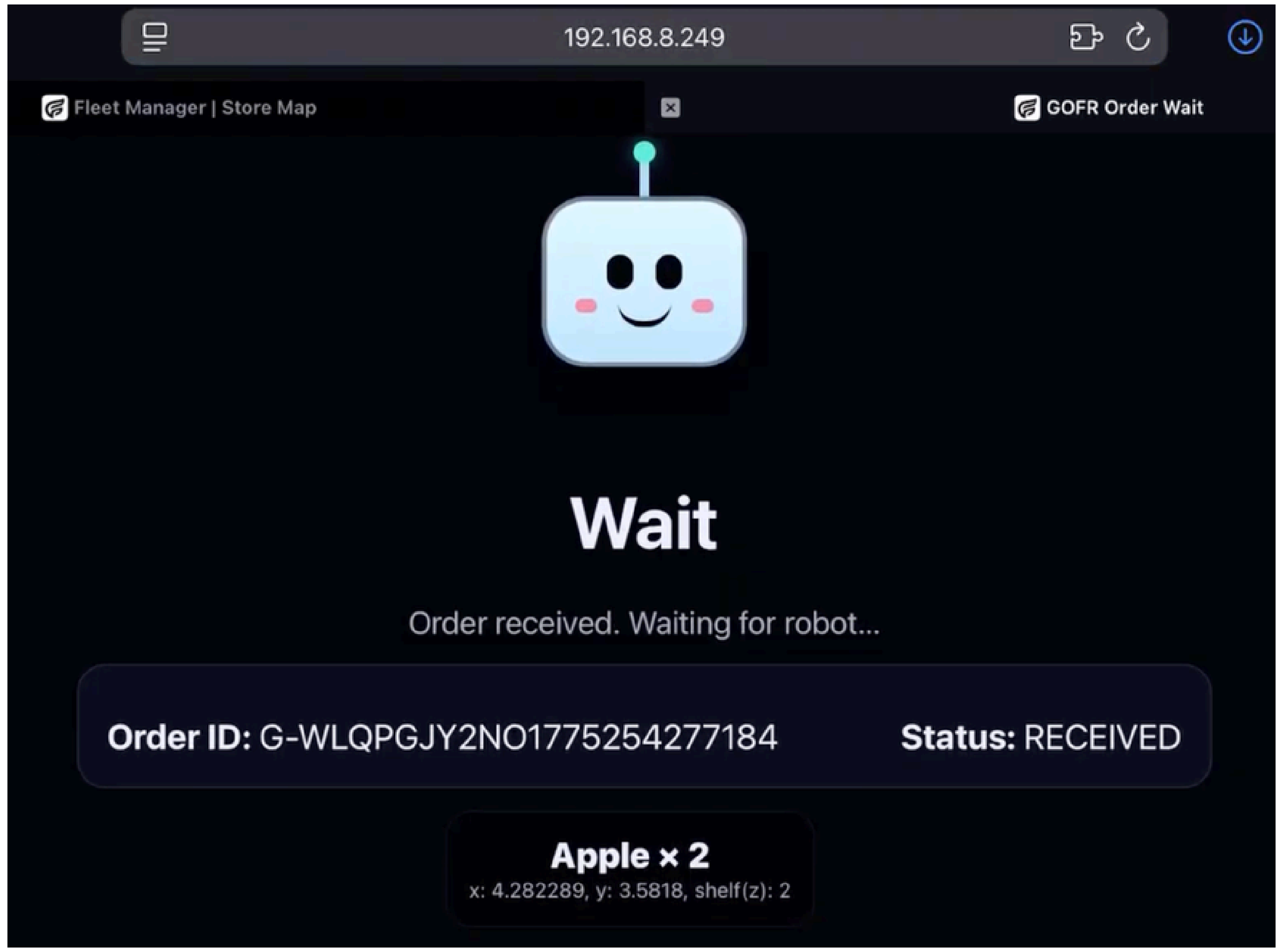
Features

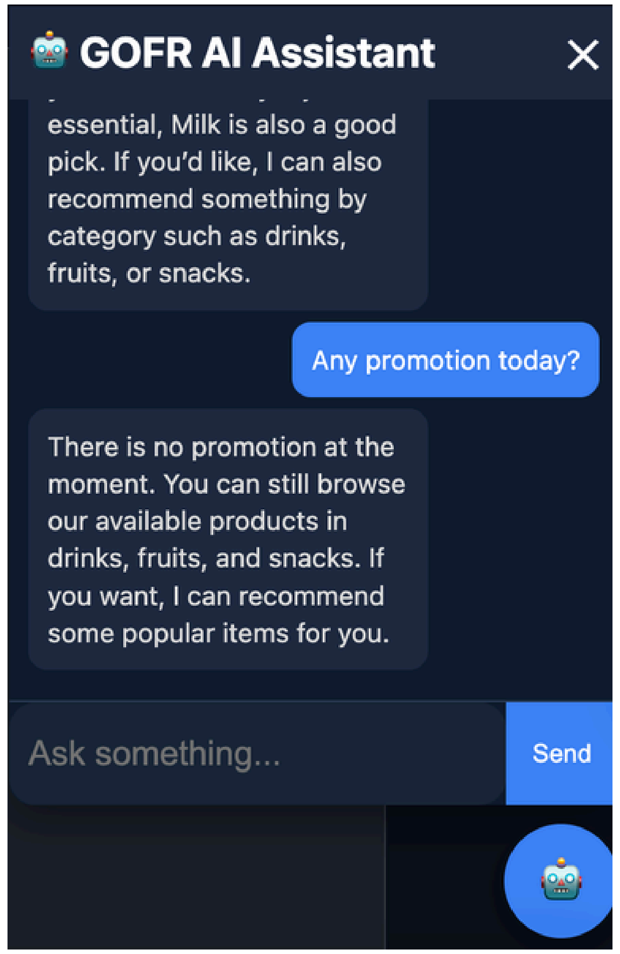
- Inventory Reports
- Store Map Visualization
- Product Location Reassignment
- Robot Status Monitoring
- **Designed for multi-robot scalability**
- AI assistant integration of Large Language Model (LLM)
- Database stored locally (Guarantee Data Privacy)





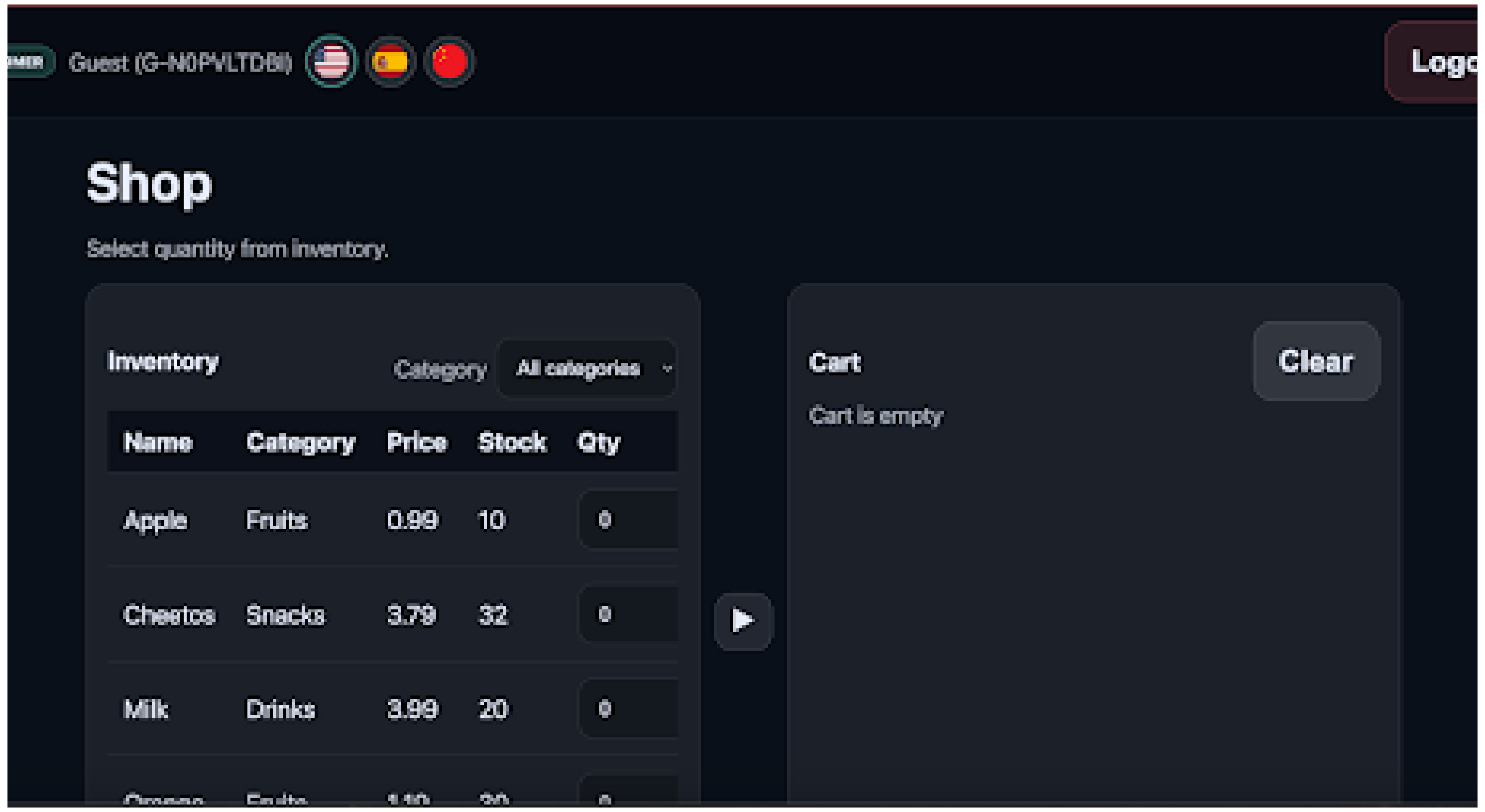
06 ONLINE ORDER APP

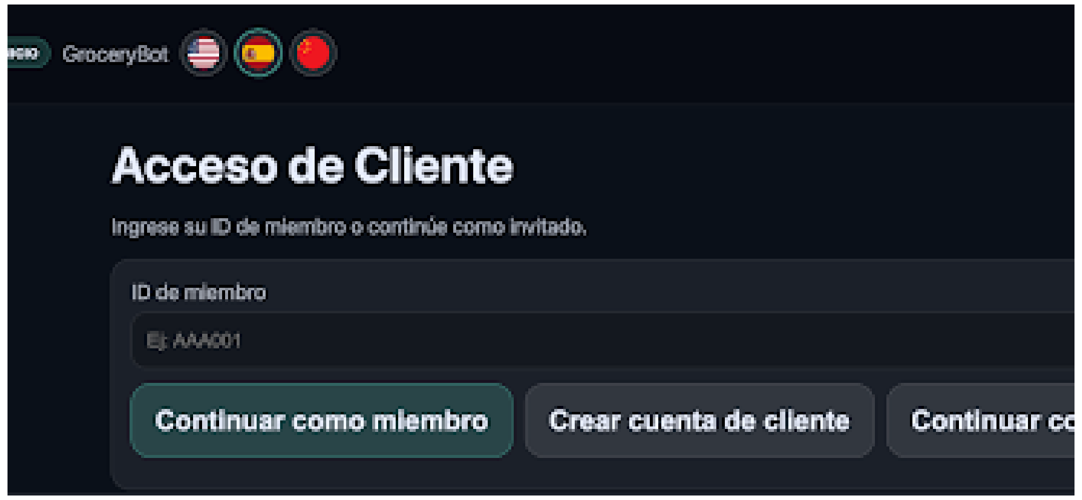




Features:

- Guest Mode
- Membership System
- Real-Time Order Tracking
- Multilingual Support (English, Spanish, Mandarin)
- AI assistant integration of Large Language Model (LLM)

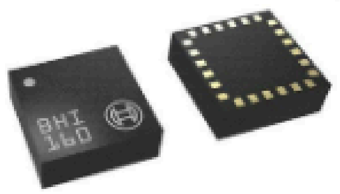


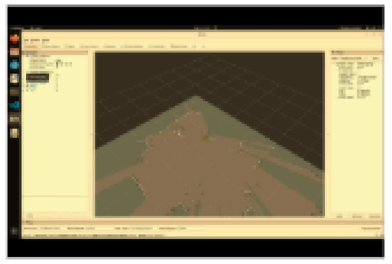


07 SYSTEM ARCHITECTURE

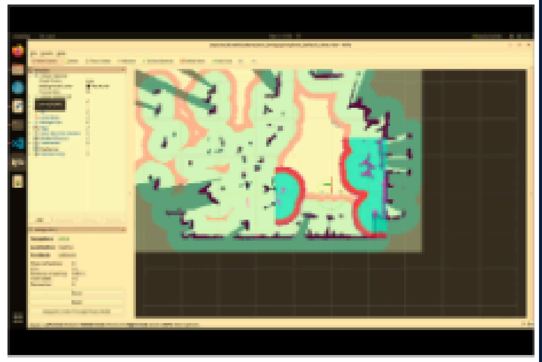
Current mainline

SICK LiDAR + IMU + odometry → **Cartographer + EKF based sensor fusion** → **Nav2 (Smac + MPPI)** → **Semantic resolve**





Reliable Navigation from SLAMed Map.

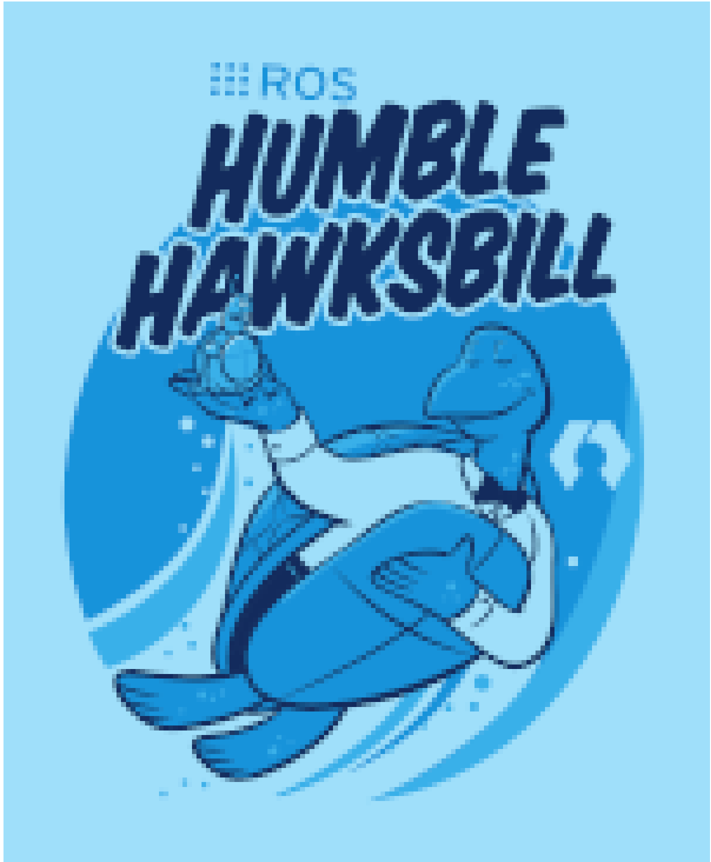


Shelves and Readable groceries on the maps

Navigation now acts as shared infrastructure for GUI, task trees, and return-home behavior.

Reusable maps → **Impact: navigation became infrastructure**

Display named targets → **Impact: navigation became infrastructure**

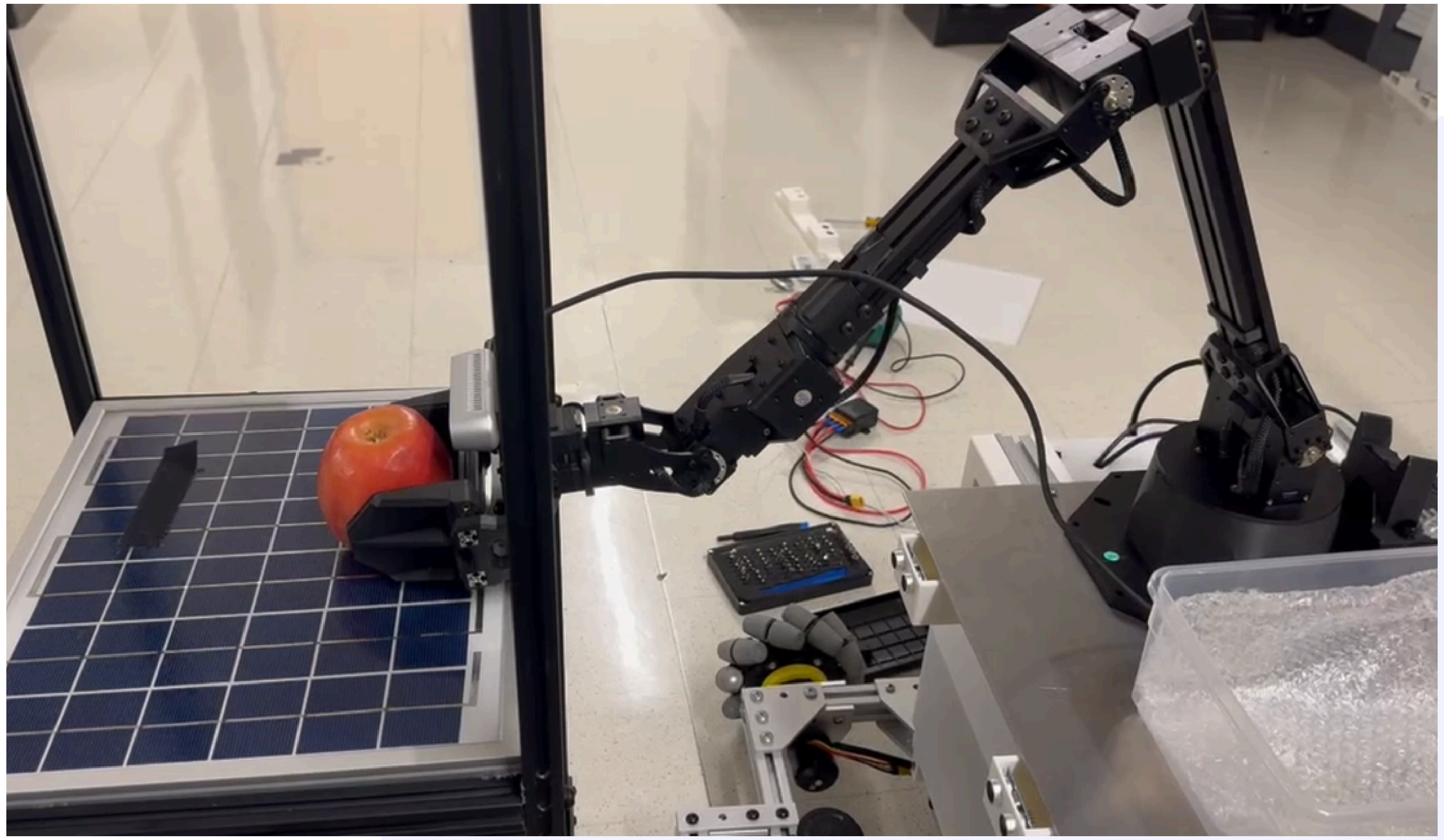


Our system uses ROS 2 with a behavior tree–based pipeline to control the robot.

Reasoning:

- ROS 2 choice: Provides modular, scalable, and reliable communication for robotics systems.
- Behavior Trees choice: Enable structured, flexible, and robust decision-making.

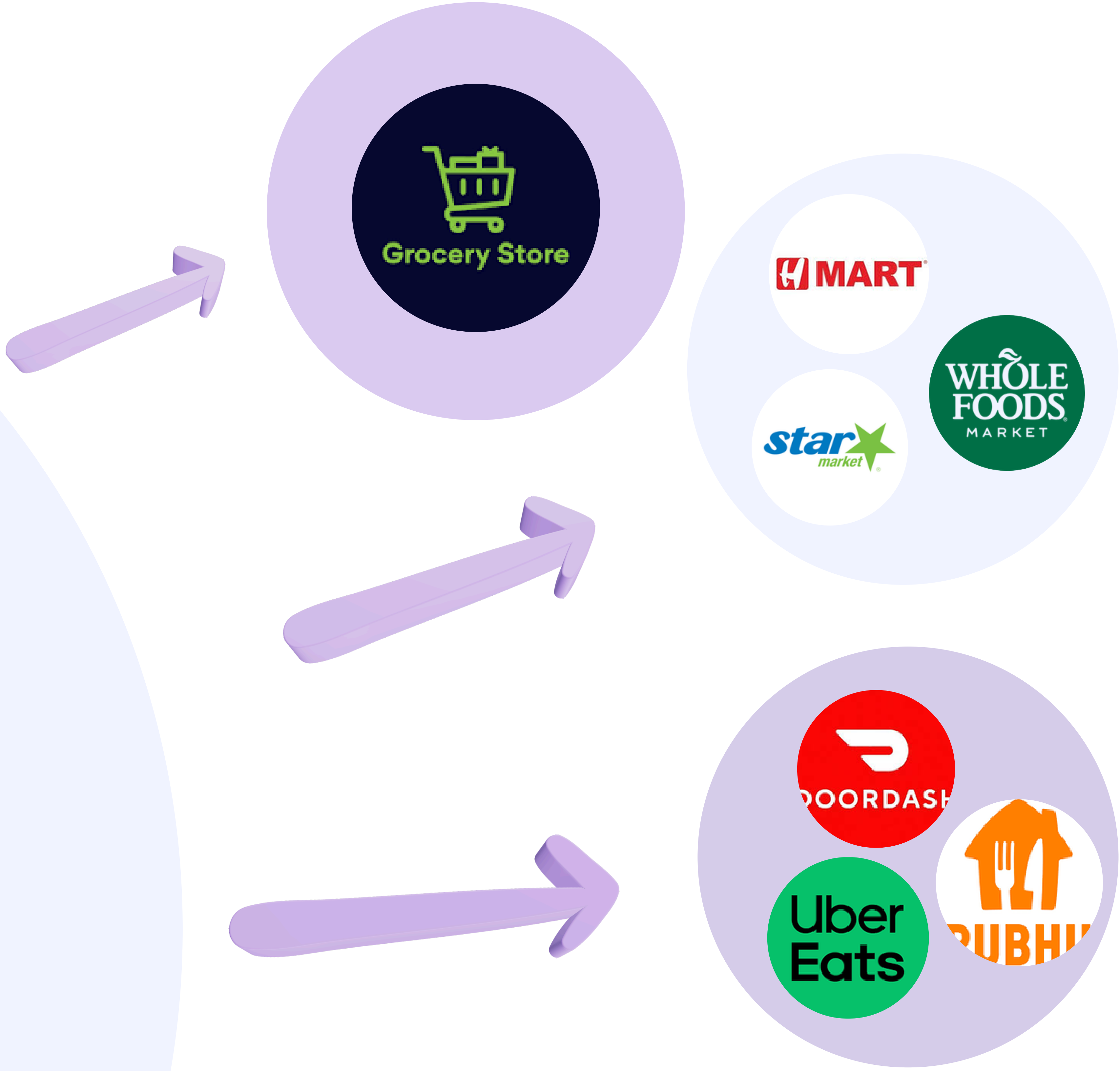
REAL-WORLD TESTING AND DEPLOYMENT



MARKET ANALYSIS

Key Sector target

1. Local Grocery Stores: Affordable robotic solutions that fit tight spaces and limited staff. Support inventory management, restocking, order fulfillment, store navigation, and customer assistance.
2. Large Supermarkets: Focus on scalability, speed, and integration with existing systems to handle heavy customer traffic and complex operations.
3. Online Delivery Platforms: Digital-first grocery services where orders are placed via apps and fulfilled before delivery personnels arrive for faster turnaround times



MARKET COMPETITORS

COMPANY	GOFR Robot	Simbe Robotics	PAL Robotics	Amazon Robotics
STRENGTH	• Dual-purpose system • LiDAR navigation • Affordable pricing	• Established brand • Inventory scanning	• Last-mile delivery • Proven technology	• Large scale automation • Sorting, picking, transport
WEAKNESS	• New market entrant	• No item manipulation • Limited to inventory	• Outdoor only • Not designed for retail stores	• Not designed for retail stores • Requires structured environments

REVENUE MODEL

ROBOT LEASING

Monthly lease per unit

STARTING AT $2,000/MO

REVENUE STREAMS:

Direct-sell

$24,000 (~40% margin)

Subscription Services (Main)

Free 1 month trial

***Detail in the next page**

Maintenance Contracts

Free Coverage over 3 years and annual checkups

Store Savings

Labor cost reduction:

70% monthly (Work double shift + cheaper than minimum wage)

$93000 per year

Market Potential

Target Market Size (US Grocery)

$900 Billion*

66,000+ stores**

5-Year Revenue Target

$110M ARR based on a 1% conversion rate within the advanced-tier package segment.

*source for market size: https://www.ibisworld.com/united-states/market-size/supermarkets-grocery-stores/1040/
**source for the number of stores: https://www.statista.com/topics/1660/food-retail/

ROBOTS AS A SERVICE (RAAS)

Package Level	Features	Price per month
Basic 1	1 robot with shopping feature only	$2000
Basic 2	1 robot with restock feature only	$2000
Advanced	1 robot with restock and shopping features	25% OFF $3000
Advanced Full Store	5 robots to operate the entire stores fully autonomous	20% OFF $12000
Delivery Service	This package targets the delivery platforms to access the API to place an order for our robot	Contact us for negotiation

Ready to Transform Grocery Operations

THANK YOU

SICK

Sensor Intelligence.

For giving us an opportunity to provide automation solution to grocery, the area people depends on everyday and sponsor Picoscan150

Connect

Partnership inquiries welcome

Github Website

https://github.com/preespp/EC463_Team_21_Grocery_Robot

THANK YOU

To sponsor our team components, funding, and provide access to a workspace to build our robotics solution

- BU College of Engineering
- BU Department of Electrical & Computer Engineering
- BU Department of Mechanical Engineering
- Robotics & Autonomous Systems Teaching and Innovation Center (RASTIC)
- Multimedia Communications Lab (Professor Thomas Little)
- Engineering Product Innovation Center (EPIC)
- Human-to-Everything (H2X) Lab (Professor Eshed Ohn-Bar)

BU

BU Electrical & Computer Engineering

BU MECHE

Electrical & Computer Engineering
Multimedia Communications Lab

BU College of Engineering
Engineering Product Innovation Center

H2X Lab

THANK YOU

For your support in providing valuable insights on commercialization models, retail sector strategies, and online delivery services. And, for helping connect with retail sector personal + providing space to test our robot

NUD POB
นัดพบ

BU Innovate@BU

- Nud Pob Thai Cuisine (Edward Chen, Owner)
- Innovate@BU (Tim Buntel, Program Director, Business Ventures)
- Anonymous Boston-Based Grocery Retail Experts (confidential due to legal considerations)
- Anonymous Boston-Area Personnel Assisting with Retail Expert Connections (confidential due to legal considerations)